

MATH 3220-002 PSet 4

Specification

Instructor: Alp Uzman

Subject to Change; Last Updated: 2025-09-19 14:18:01-06:00

1 Background

This problem set focuses on the topological structure of metric spaces. Recall that a metric space is nothing but a set together with a distance function satisfying certain axioms, modeling the notion of "distance between two points". As such, metric spaces are very versatile and common objects; in particular many sets of functions can be considered as metric spaces (possibly in multiple, non-equivalent ways).

From a topological point of view, within a metric space three types of subsets are of paramount importance:

- **Open subsets** are those that are closed under sufficiently small perturbations. Interior operation is the set operation that transforms any set into an open set optimally.
- **Closed subsets** are those that are closed under limits. Closure operation is the set operation that transforms any set into a closed set optimally.
- **Compact subsets** are those in which any sequence converges to some point in the subset up to a subsequence.

First two problems explore open and closed sets. Problem 3 then focuses on compactness, and finally in Problem 4 develops the understanding of function spaces as metric spaces.

2 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details. The [hyperlinks](#) too are meant to serve as optional additional context; chasing and studying them are by no means necessary to solve the problems. Indeed this problem set is designed so that, broadly speaking, the lectures are sufficient to complete it.

When tackling statements in MATH 3220, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.
2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While

work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical **error detection and correction skills**.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example (say as proof to a given existential statement), you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook; it's up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. Let X be a metric space. For any $S \subseteq X$, abbreviate the closure and interior of S respectively by κS and ιS . Let $A, B \subseteq X$. Then

(a) $\iota A \subseteq A \subseteq \kappa A$.

- (b) A is open iff $\iota A = A$.
- (c) A is closed iff $A = \kappa A$.
- (d) $\iota \iota A = \iota A$.
- (e) $\kappa \kappa A = \kappa A$.
- (f) $A \subseteq B$ iff $\iota A \subseteq \iota B$.
- (g) $A \subseteq B$ iff $\kappa A \subseteq \kappa B$.
- (h) A is open iff $A \subseteq \iota \kappa A$.
- (i) A is closed iff $A \supseteq \kappa \iota A$.
- (j) $\iota A \subseteq \iota \kappa \iota A \subseteq \kappa \iota \kappa A \subseteq \kappa A$.
- (k) $\iota A \subseteq \iota \kappa A \subseteq \kappa \iota \kappa A \subseteq \kappa A$.
- (l) $\iota \kappa \iota \kappa A = \iota \kappa A$.
- (m) $\kappa \iota \kappa \iota A = \kappa \iota A$.
- (n) One can not obtain more than 7 distinct subsets by taking interiors and closures of a subset A of X iteratively, that is, the following collection of subsets can not have more than 7 elements:

$$\{A, \\ \iota A, \kappa A, \\ \kappa \iota A, \iota \kappa A, \\ \iota \kappa \iota A, \kappa \iota \kappa A, \\ \kappa \iota \kappa \iota A, \iota \kappa \iota \kappa A, \\ \iota \kappa \iota \kappa \iota A, \kappa \iota \kappa \iota \kappa A, \\ \kappa \iota \kappa \iota \kappa \iota A, \iota \kappa \iota \kappa \iota \kappa A, \dots\}$$

- (o) There is a set R of real numbers such that each of the following subsets is distinct:

$$R, \iota R, \kappa R, \kappa \iota R, \iota \kappa R, \iota \kappa \iota R, \kappa \iota \kappa R.$$

2. Let X be a metric space and $A, B \subseteq X$. Then

(a) $\text{clos}(A \cup B) = \text{clos}(A) \cup \text{clos}(B)$.

(b) $\text{clos}(A \cap B) = \text{clos}(A) \cap \text{clos}(B)$.

(c) $\text{int}(A \cup B) = \text{int}(A) \cup \text{int}(B)$.

(d) $\text{int}(A \cap B) = \text{int}(A) \cap \text{int}(B)$.

3. Let X be a metric space, $A \subseteq X$. Then the following are equivalent:

(a) A is compact (that is, any sequence in A has a subsequence that converges to some point in A).

(b) for any metric space Y , $C^0(A; Y) \subseteq F_b(A; Y)$.

(c) for any metric space Y and for any $f \in C^0(A; Y)$, $f(A)$ is a compact subset of Y .

(d) for any $f \in C_b^0(A; \mathbb{R})$, f attains its maximum and minimum.

(e) For any decreasing sequence $F_\bullet$ of closed subsets of A , the intersection $\bigcap_n F_n$ is nonempty.

4. Consider the following sequences $f_\bullet$ of functions defined on $[0, 1]^3$:

I: $n \mapsto \left[(x, y, z) \mapsto \left(\frac{x}{1 + nx}, \frac{y}{n} \right) \right]$.

II: $n \mapsto \left[(x, y, z) \mapsto \left(\frac{x}{1 + ny}, \frac{yz}{n} \right) \right]$.

III: $n \mapsto \left[(x, y, z) \mapsto \left(\frac{\sin(2\pi nx)}{n}, \cos(2\pi y/n), (yz)^n \right) \right]$.

For each sequence $f_\bullet$, perform the following tasks:

- (a) Determine whether or not $f_\bullet$ is pointwise convergent.
- (b) If $f_\bullet$ is pointwise convergent, compute its pointwise limit.
- (c) Determine whether or not $f_\bullet$ is uniformly convergent.
- (d) If $f_\bullet$ is uniformly convergent, compute its uniform limit.
- (e) Determine whether or not $f_\bullet$ is a bounded sequence.
- (f) Determine a pointwise convergent subsequence of $f_\bullet$ and compute the pointwise limit of this subsequence.
- (g) Determine a uniformly convergent subsequence of $f_\bullet$ and compute the uniform limit of this subsequence.

3 Generative AI and Computer Algebra Systems Regulations

This section applies if you decide to use either a **generative AI tool** or a **computer algebra system** for this problem set. If not, you may skip this section.

3.1 Submission Requirements: Interaction Logs

If you use such tools, you must submit a complete log of your interactions. This is a mandatory part of your submission.

You can fulfill this requirement by either:

- **Listing the URLs** for your interactions in the appropriate field of the submission form, or
- **Attaching PDFs** containing the full interaction history to your Gradescope submission.

How to Obtain Logs You can generate the necessary logs using one of the following methods:

- **Shared Links:** Some AI tools (e.g., **ChatGPT**) can generate a unique, shareable URL for a conversation.
- **Web Archives:** Use a service like the **Wayback Machine** (see their "**Save Page Now**" **documentation**) to create a permanent, timestamped archive of your chat. There are also third-party browser extensions that are designed to extract chat logs from generative tools.
- **Screenshots:** Manually take screenshots of the entire interaction and compile them into a single PDF file. You can use your operating system's built-in tools or a third-party application.

3.2 Chat Guidelines and Prompt Engineering

When using chatbots, you must adhere to the following guidelines:

- **Permitted:** You may copy and paste text from this problem set, the textbook, or other course materials into the chat to provide context.
- **Prohibited:** You may not directly ask for a complete solution to a problem.
- **Required:** You must begin your conversation with a "guardrails" prompt designed to structure the AI's responses. This practice is known as **prompt engineering**, and it is your responsibility you design an appropriate guardrails prompt. The staff will evaluate the reasonableness of your attempts based on the logs you submit.

Here are some guides regarding prompt engineering:

- <https://platform.openai.com/docs/guides/prompt-engineering>
- <https://developers.google.com/machine-learning/resources/prompt-eng>
- <https://www.ibm.com/topics/prompt-engineering>
- <https://aws.amazon.com/what-is/prompt-engineering/>

When in doubt, check with the course staff to ensure your approach is appropriate.

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

<https://forms.gle/7r8Rb8ZbHrbYbxQz9>.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

<https://www.gradescope.com/courses/1073533/assignments/6451768>,

see the Gradescope [documentation](#) for instructions.

You will receive a zero for this assignment if you submit your work on Gradescope under the wrong assignment, even if you submit your work on time.

5 When to Submit

This problem set is due on September 26, 2025 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.